

T3 analytic one-loop group factors

This report presents the validated analytic representation-dependent group factors used by the T3 RGE pipeline. Model dimensions are read from `common.T3Model.T3_CLASSES` and model-specific coefficients are evaluated from the analytic group-factor implementations, not copied from RGBeta comparison tables.

Representation conventions

$$j = \frac{d-1}{2}, \quad C_2(d) = \frac{d^2-1}{4}, \quad T(d) = \frac{d(d^2-1)}{12}.$$

For each Yukawa vertex define

$$G_S = \frac{d_{>}}{d_S}, \quad G_L = \frac{d_{>}}{2}, \quad G_F = \frac{d_{>}}{d_F}, \quad d_{>} = \max(d_S, d_F).$$

Yukawa beta functions

$$\begin{aligned} 16\pi^2\beta_{y_1} &= G_{S_1} \text{Tr}(y_1 y_1^\dagger) y_1 + \frac{1}{2}(G_{L_1} + G_{F_1}) y_1 y_1^\dagger y_1 \\ &\quad + \frac{1}{2} G_{L_2} y_2 y_2^\dagger y_1 + \frac{1}{2} Y_e Y_e^\dagger y_1 \\ &\quad - 3g_2^2 [C_2(L) + C_2(F)] y_1 - 3g_Y^2 [Y_L^2 + Y_F^2] y_1. \\ 16\pi^2\beta_{y_2} &= G_{S_2} \text{Tr}(y_2 y_2^\dagger) y_2 + \frac{1}{2}(G_{L_2} + G_{F_2}) y_2 y_2^\dagger y_2 \\ &\quad + \frac{1}{2} G_{L_1} y_1 y_1^\dagger y_2 + \frac{1}{2} Y_e Y_e^\dagger y_2 \\ &\quad - 3g_2^2 [C_2(L) + C_2(F)] y_2 - 3g_Y^2 [Y_L^2 + Y_F^2] y_2 \\ &\quad + \delta_{\text{SC}} \left[G_{S_2} \text{Tr}(y_2 y_1^\dagger) y_1 + \frac{1}{2} G_{F_2} y_2 y_1^\dagger y_1 \right]. \end{aligned}$$

Here $\delta_{\text{SC}} = 1$ only for the physical self-conjugate branch $Y_F = 0$ with odd d_F ; otherwise it is zero.

Fermion mass

$$16\pi^2\beta_{M_F} = \frac{1}{2} G_{F_1} (y_1^T y_1^*) M_F + \frac{1}{2} G_{F_2} M_F (y_2^\dagger y_2) - 6g_2^2 C_2(F) M_F - 6g_Y^2 Y_F^2 M_F.$$

Heavy scalar masses

$$\begin{aligned} 16\pi^2\beta_{m_{S_1}^2} &= 2G_{S_1} m_{S_1}^2 \text{Tr}(y_1 y_1^\dagger) + \Xi_F G_{S_1} \text{Tr}(M_F M_F^\dagger y_1^T y_1^*) \\ &\quad + 2(d_{S_1} + 1) \lambda_{S_1} m_{S_1}^2 - 6C_2(S_1) g_2^2 m_{S_1}^2 - 6Y_{S_1}^2 g_Y^2 m_{S_1}^2 \\ &\quad + 2d_{S_2} \lambda_{12} m_{S_2}^2 + \dots, \end{aligned}$$

with $\Xi_F = -4$ on the vector-like branch and $\Xi_F = -16$ on the physical self-conjugate branch. The S_2 equation follows by $1 \leftrightarrow 2$.

Direct Weinberg mixing

$$\begin{aligned} 16\pi^2\beta_\kappa &= R_W \lambda_{T3} C_{12}, \\ R_W &= \frac{4}{3} \eta_F \sqrt{3d_{S_1} d_{S_2} d_F} \begin{Bmatrix} \frac{1}{2} & \frac{1}{2} & 1 \\ j_{S_2} & j_{S_1} & j_F \end{Bmatrix}, \quad \eta_F = (+1, +1, -1) \text{ for } d_F = (1, 2, 3). \end{aligned}$$

Representation summary

Model	d_{S_1}	d_{S_2}	d_F	G_{S_1}	c_{11}	c_{21}	G_{S_2}	c_{22}	c_{12}	R_W
T3-A	1	3	2	2	1	$\frac{3}{4}$	1	$\frac{3}{2}$	$\frac{1}{2}$	$\frac{4\sqrt{3}}{3}$
T3-B	2	2	1	1	$\frac{3}{2}$	$\frac{1}{2}$	1	$\frac{3}{2}$	$\frac{1}{2}$	$\frac{4\sqrt{3}}{3}$
T3-C	2	2	3	$\frac{3}{2}$	$\frac{5}{4}$	$\frac{3}{4}$	$\frac{3}{2}$	$\frac{5}{4}$	$\frac{3}{4}$	$-\frac{4}{3}$
T3-D	3	1	2	1	$\frac{3}{2}$	$\frac{1}{2}$	2	1	$\frac{3}{4}$	$\frac{4\sqrt{3}}{3}$
T3-E	3	3	2	1	$\frac{3}{2}$	$\frac{3}{4}$	1	$\frac{3}{2}$	$\frac{3}{4}$	$-\frac{4\sqrt{6}}{3}$

$$c_{11} = \frac{1}{2}(G_{L_1} + G_{F_1}), c_{21} = \frac{1}{2}G_{L_2}, c_{22} = \frac{1}{2}(G_{L_2} + G_{F_2}), c_{12} = \frac{1}{2}G_{L_1}.$$

Scalar-mass coefficient check at $\alpha = 0$

Model	$2G_{S_1}$	$-4G_{S_1}$	$2(d_{S_1} + 1)$	$2d_{S_2}$	$2G_{S_2}$	$-4G_{S_2}$	$2(d_{S_2} + 1)$	$2d_{S_1}$
T3-A	4	-8	4	6	2	-4	8	2
T3-B	2	-4	6	4	2	-4	6	4
T3-C	3	-6	6	4	3	-6	6	4
T3-D	2	-4	8	2	4	-8	4	6
T3-E	2	-4	8	6	2	-4	8	6